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A new optimized fuzzy FOPI-FOPD controller for automatic generation control of electric power systems

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Abstract

In the rapidly expanding size and complexity of the electricity network, automatic generation control (AGC) is contemplated to be the most remarkable option for offering good quality electric power supply to the end users. An AGC system entails highly vigorous, competent and intelligent control technique to deliver a healthy power under stochastic nature of consumers' power demand. Hence, in this paper, a hybrid fuzzy fractional order proportional integral-fractional order proportional derivative (FFOPI-FOPD) controller is proposed as a new expert control technique to tackle AGC profitably in isolated and interconnected multi-area power systems. A recently developed imperialist competitive algorithm (ICA) is utilized for the optimization of the output gains ($K_P/K_{P1}/K_I/K_D$) and other parameters such as order of integrator (λ) and differentiator (μ) of FFOPI-FOPD controller exercising integral of squared error criterion. The proposed technique is firstly implemented on 1-area thermal system, then to express its potential and extensibility, the work is extended to 2-area hydro-thermal and 3-area thermal power systems widespread in the literature. The eminence of the method is betokened by comparing the results with the various newly published control methodologies and FPI/FFOPI controller designed in the study via ICA in terms of minimum values of various error criteria and undershoots/overshoots/settling times of frequency and tie-line power deviations following a sudden load demand in an area. The sensitivity analysis substantiates that the suggested controller is robust and performs staunchly under the wide variations in the system parameters, random load pattern and in the company of physical constraints to produce more clean electricity.
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1. Introduction

The foremost intentions of the power system utility include generating, transmitting, distributing and controlling of electrical power economically and competently while preserving an uninterrupted supply of the electricity within a tolerable quality to the consumers. Automatic generation control (AGC) is a crucial issue for triumphant operation of power system to deliver quality power [1]. The chief intention of AGC is to ensure the deviation in frequency and the tie-line power flows within required limits in the wake of consumer load requirements and various disturbances via maintaining the balance between generation and power demand plus losses [2]. In view of it, an advanced intelligent AGC technique is necessitated to nullify the impacts of random load variations for retaining the frequency, generation and tie-line flows at nominal value.

Hence, researchers are endeavouring to propose new control structures for AGC of power systems. The literature study discloses that several development and research strategies have been introduced such as classical optimal PI [1,2], genetic algorithm (GA) based classical I/PI/PID/fractional order PID (FOPID) [3,4], blended biogeography based optimization (BBBO) based PI [5], Grey wolves-based optimization (GWO) based PID [6], krill herd algorithm (KHA) based I/PI/PID/IDD/PIDD [7], harmony search algorithm (HSA) based PID [8], quasi-oppositional differential search algorithm (QODSA) based PI/PID [9], hybrid firefly algorithm and pattern search (hFA-PS) based PI/PID [10], hybrid adaptive gbest-guided gravitational search and pattern search algorithm (hGGSA-PS) based PI/PID [11], bacterial foraging optimization algorithm (BFOA) based I [12] and ICA based PID [13,14] controllers for varied types of power system models. It is observed that the outcome of AGC system largely depends on the structure of the controller and the artificial intelligent techniques employed to tune the controller parameters. Consequently, suggesting new high performance optimization algorithms with new control designs to AGC problem is always appreciated. However, classical controllers show limitations under changed operating conditions and less robustness under disturbances and system uncertainties.

To overcome the problems of classical controllers, recently, neural network (NN) control due to its good performance in uncertain environment, nonlinear systems and its capability of universal approximation is enchanting the researchers [15–18]. A generalised Hopfield NN based self-adaptive PID controller for AGC is designed for a nonlinear 2-area power system [15]. Additionally, NN control also successfully deals with the uncertainties in complex nonlinear flapping wing micro aerial vehicle dynamic system utilizing full state and output feedback strategy [16]. In [17], a NN controller effectively arrests the vibrations in a flexible robotic manipulator system having input deadzone via approximating its dynamics. An adaptive fuzzy NN control is proposed for an erratic constrained robot with unknown dynamics and state constraints in [18]. These NN methods display good dynamical outcomes in comparison to classical control methods but some of them advocate state feedback which is not practically viable for power industry practices. In a large-scale power system, the total number of system states may be much large and many of them are not accessible. However, reconstruction of states may be done but it comprises extra cost and intricacy in the execution.

Currently, fuzzy logic controller (FLC) is employed in almost all sectors of industry and science such as robotics [18], complex nonlinear Markov jump systems [19], nonlinear singularly perturbed systems [20] and power system operation and control [21–38]. The literature survey indicates that FLC refines the closed loop potential of classical controller and can effectively tackle the variations in operating point or system parameters via online correc-

tion of controller parameters. FLC performance depends mainly on the proper selection of fuzzy parameters like input/output scaling factors (SFs), membership functions, rule base, defuzzification method etc. Usually, they are chosen via some specific empirical rules and hence, may not be the optimal ones. However, their inappropriate choice may influence the performance of a FLC substantially. In the recent times, several researchers have applied diverse new techniques and structures like hybrid particle swarm optimization-pattern search (hPSO-PS) [21]/ICA [22]/BFOA [23] to optimize fuzzy PI (FPI) and BFOA [23]/hybrid differential evolution-GWO (hDE-GWO) algorithm [24]/PSO [25]/sine cosine algorithm (SCA) [26]/GWO algorithm [27,28] to optimize fuzzy PID (FPID) controllers to upgrade the dynamic performance of FPI/FPID/FPID with filter (FPIDF) controllers. The optimized FLC offers superior performance compared to unoptimized fuzzy and optimized classical controllers. Recently proposed ICA optimized FPIDF-(1+PI) [29] and FPIDF plus double integral (FPIDF-II) [30] controllers display worth-appreciating performance in AGC of power systems.

In the recent years, an emerging temptation in the hybridization of fractional calculus with FLC is observed. This practice widens the scope to escalate the AGC system control performance of FLC. An ICA optimized FPIDF plus fractional order integral (FOI) controller is proposed for AGC of various power system models in [31]. FO fuzzy PID (FOFPID) controllers optimized via biogeography-based optimization (BBO) [32]/cuckoo search algorithm (CSA) [33]/BFOA [34,35]/ICA [36]/chaotic PSO [37]/quasi-oppositional harmony search algorithm (QOHSA) [38] are proposed for different traditional and restructured power systems. The system results due to FPIDF-FOI/FOFPID are found better compared to FOPID/FPIDF/FPID controller [31–38].

The extant optimization methods implemented to AGC problem show deficiency to offer a convenient adjustment between their explorative and exploitative talent, hence, they have tendency to trap into local optima. In the recent past, ICA is widely endorsed for optimization of various control strategies by reason of its potential of handling parameter ambiguity, robustness, global search and stable convergence character [13,14,22,29–31,36]. ICA is inspired by the imperialistic competition and socio-political evolution of humans [36]. The versatility, superior performance and apropos convergence and global minimum attainment of ICA motivated to analyze its application to optimize a new controller proposed in the present work for AGC study of power systems.

The literature survey acknowledges that various fuzzy control structures with different optimization techniques have been advised for AGC of power systems such as FPI/FPID/FPIDF/FPIDF-(1+PI)/FPIDF-II [21–30] and FPIDF-FOI/FOFPID [31–38]. Next, optimized FOI with 2 degree of freedom-PID with filter (2DOF-PIDF-FOI) [39], FOPI [40] and FOPD with PIDF (PIDF-FOPD) [41] controllers are also observed lately. But, no attempt has been made so far to design FOPD controller with fuzzy FOPI i.e., FFOPI-FOPD structured AGC controller for power system. Hence, motivated from the above developments, in this paper, a new FFOPI-FOPD controller is proposed for AGC of single/multi-area power system and ICA is implemented to tune controller parameters like K_P, K_I, λ, K_{P1}, K_D and μ such that it minimizes the oscillations, overshoots, undershoots and settling time in the system responses under sudden load deviations, thus refining the power system AGC performance notably. The prominent significances of the presented work are summarized as follows:

(a) To design and implement a new FFOPI-FOPD controller for AGC problem solution of 1-area non-reheat thermal, 2-area hydro-thermal and 3-area reheat thermal power systems.

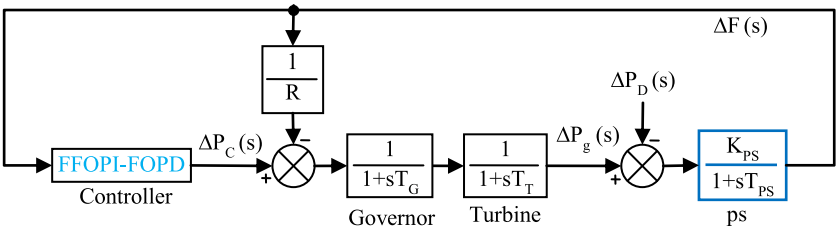


Fig. 1. Transfer function model of 1-area non-reheat thermal power system [4,8].

(b) To tune proposed controller parameters using ICA.
(c) To demonstrate the advantage of the proposed ICA tuned FFOPI-FOPD controller over optimal/GA/KHA/QODSA based PI, GA/HS/BBBO/hFA-PS/hGGSA-PS/GWO/KHA based PID, GA based FOPID and ICA based FPI/FFOPI controllers.
(d) To conduct sensitivity analysis to confirm the robustness of the proposed controller under parametric variations, random load demands and physical constraints.
(e) To study the impact of the physical constraints like generation rate constraint, governor dead band nonlinearity and time delay on the system dynamic performance under above point (d).

The paper is arranged as follows: following a comprehensive literature review in Section 1, the description of the models under study is provided in Section 2. In Section 3, the structure of FFOPI-FOPD controller is presented. In Section 4, performance index utilized is defined. In Section 5, the simulation results are explained in detail. In Section 6, sensitivity analysis of the proposed controlled is demonstrated. Finally, conclusion and future scope of the work are pointed out in Section 7.

2. Power system models

The investigated systems under this study are 1-area non-reheat thermal, 2-area interconnected hydro-thermal and 3-area interconnected reheat thermal power systems. The modelling of the systems is available in the literature for 1-area non-reheat thermal power system (Fig. 1) [4], 2-area hydro-thermal power system (Fig. 2) [3] and 3-area reheat thermal power system (Fig. 3) [2,7]. In the systems under study, 1-area system contains 1 non-reheat thermal unit. Each control area of 2-area system owns 1 non-reheat thermal and 1 mechanical governor based hydro units and every control area of 3-area system has one reheat thermal unit. The control area ratings of 1/2-area system is 2000 MW and 1/2/3-area of 3-area system is 2000 MW/5000 MW/8000 MW, respectively. The nominal parameters and nomenclature displayed in the figures are elucidated in Appendix.

3. FFOPI-FOPD controller structure

A cascade combination of fuzzy, FOPI and FOPD controllers called FFOPI-FOPD controller is proposed for the first time in AGC. The FOPI (or PI^{λ}) and FOPD (or PD^{μ}) controllers are generalizations of PI and PD controllers, respectively, where the orders of the integral and the derivative are not integers [40,41]. This attribute offers more pliability to the controller structure to lend superior dynamic performance of the system. The continuous transfer

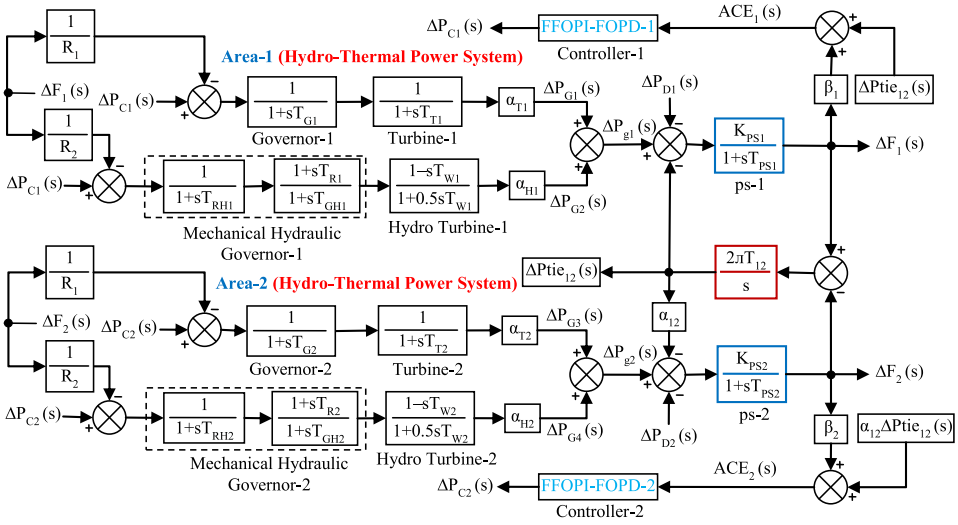


Fig. 2. Transfer function model of 2-area hydro-thermal power system [3].

functions of FOPI and FOPD controllers are stated as:

$$\text{FOPI} = K_P + K_I s^{-\lambda} \tag{1}$$

$$\text{FOPD} = K_{P1} + K_D s^{\mu} \tag{2}$$

where λ, $\mu > 0$. The K_P, K_I, K_{P1} and K_D are the proportional, integral, proportional and derivative gains, respectively. Hence, the controller has six autonomous tuning knobs. The optimal values of these six gains are determined employing ICA. For $\lambda = \mu = 1$, the controller structure of FOPI-FOPD Eqs. (1) and (2) reduces to the classical PI-PD controller. The structure of the proposed controller is shown in Fig. 4(a) in which ACE (area control error) and derivative of ACE are the inputs to the fuzzy logic controller (FLC), y is the output of FLC and it is also the input to the cascaded FOPI-FOPD controller, and ΔP_C is the output from the suggested FFOPI-FOPD controller. The output/control signal of the controller is given by:

$$\Delta P_C = y \times (K_P + K_I s^{-\lambda}) \times (K_{P1} + K_D s^{\mu}) \tag{3}$$

The ACE for 1, 2 and 3-area systems is stated as follows:

$$\text{ACE} = \Delta F \,(1-\text{area system}) \tag{4}$$

$$\left.\begin{aligned} \text{ACE}_1 &= \beta_1 \Delta F_1 + \Delta\text{Ptie}_{12} \\ \text{ACE}_2 &= \beta_2 \Delta F_2 + \alpha_{12} \Delta\text{Ptie}_{12} \end{aligned}\right\} \ (2-\text{area system}) \tag{5}$$

$$\left.\begin{aligned} \text{ACE}_1 &= \beta_1 \Delta F_1 + \Delta\text{Ptie}_1 \\ \text{ACE}_2 &= \beta_2 \Delta F_2 + \Delta\text{Ptie}_2 \\ \text{ACE}_3 &= \beta_3 \Delta F_3 + \Delta\text{Ptie}_3 \end{aligned}\right\} \ (3-\text{area system}) \tag{6}$$

where β_i is the frequency response characteristic, ΔF_i is the deviation in control area frequency, $\Delta\text{Ptie}_{ij}/\Delta\text{Ptie}_i$ is the deviation in tie-line power flow, α_{ij} is the control area

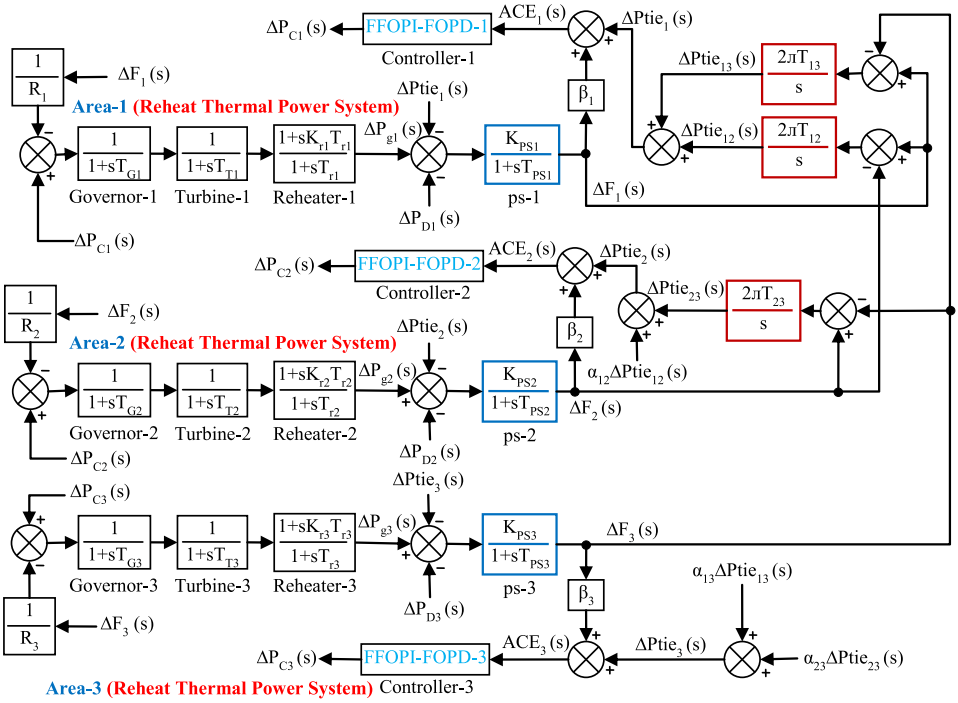


Fig. 3. Transfer function model of 3-area reheat thermal power system [2,7].

Table 1
Rule base for ACE, ACE derivative and FLC output [29–31].

ACE	ACE derivative						
	LN	MN	SN	Z	SP	MP	LP
LN	LP	LP	LP	MP	MP	SP	Z
MN	LP	MP	MP	MP	SP	Z	SN
SN	LP	MP	SP	SP	Z	SN	MN
Z	MP	MP	SP	Z	SN	MN	MN
SP	MP	SP	Z	SN	SN	MN	LN
MP	SP	Z	SN	MN	MN	MN	LN
LP	Z	SN	MN	MN	LN	LN	LN

size ratio and $\Delta Ptie_1 = \Delta Ptie_{12} + \Delta Ptie_{13}$, $\Delta Ptie_2 = \alpha_{12}\Delta Ptie_{12} + \Delta Ptie_{23}$ and $\Delta Ptie_3 = \alpha_{13}\Delta Ptie_{13} + \alpha_{23}\Delta Ptie_{23}$.

For FLCs, optimizing the shape of membership functions (MFs) exhibits less influence on the outcome of the closed loop system than that of its input/output scaling factors (SFs) [21–38]. Hence, in the present work, stress is given on supposing the MFs and rule base for FLC as invariant ones [21–26,28–32,34–38] and only output SFs are optimized [22,29–31]. The fuzzy rule base engaged in FFOPI-FOPD controller is depicted in Table 1 [29–31]. However, the fixed shaped MFs for inputs and FLC output are presented in Fig. 4(b) [22,29–31]. The fuzzy linguistic variables shown in Fig. 4(b) such as LN, MN, SN, Z, SP, MP and

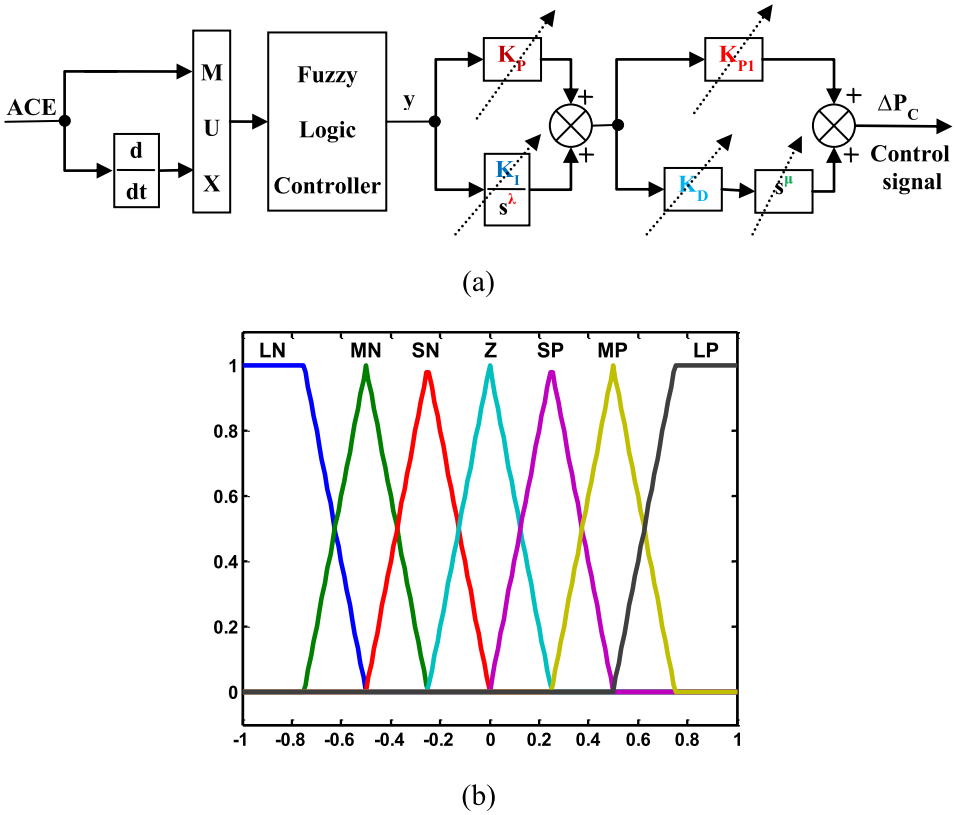


Fig. 4. Proposed FFOPI-FOPD controller. (a) Structure and (b) membership functions for inputs/output [22,29–31].

LP are symbolized as large negative, medium negative, small negative, zero, small positive, medium positive and large positive, respectively. Mamdani fuzzy inference system and center of gravity method of defuzzification are used to get real control output of FLC.

4. Performance index

For success of controlling action from a controller, its parameters necessitate to be optimized. For non-linear controller like FLC, there is not any clear-cut optimization rule. That is why, time integral performance indices like integral of absolute error (IAE), integral of time multiplied absolute error (ITAE), integral of squared error (ISE) and integral of time multiplied squared error (ITSE) are of highly encouraging. These time-domain performance indices (J_S) for 2-area power system are stated by the following expressions:

$$\text{IAE} = \int_0^\infty \{|\Delta F_1| + |\Delta F_2| + |\Delta \text{Ptie}_{12}|\}\, \text{dt} \tag{7}$$

$$\text{ITAE} = \int_0^\infty t\{|\Delta F_1| + |\Delta F_2| + |\Delta \text{Ptie}_{12}|\}\text{dt} \tag{8}$$

$$\text{ISE} = \int_0^\infty \left\{\Delta F_1^2 + \Delta F_2^2 + \Delta \text{Ptie}_{12}^2\right\} \text{dt} \tag{9}$$

$$\text{ITSE} = \int_0^\infty t\left\{\Delta F_1^2 + \Delta F_2^2 + \Delta \text{Ptie}_{12}^2\right\} \text{dt} \tag{10}$$

The J_S for 1 and 3-area systems can be scripted correspondingly. In the current work, the controllers are optimized employing ISE as a J, subjected to the limits of six tunable variables. Consequently, the optimization problem may be devised as a constrained optimization problem. Minimize J (= ISE) subjected to:

$$\left.\begin{array}{l} K_P^{\min} \le K_P \le K_P^{\max} \\ K_I^{\min} \le K_I \le K_I^{\max} \\ \lambda^{\min} \le \lambda \le \lambda^{\max} \\ K_{P1}^{\min} \le K_{P1} \le K_{P1}^{\max} \\ K_D^{\min} \le K_D \le K_D^{\max} \\ \mu^{\min} \le \mu \le \mu^{\max} \end{array}\right\} \tag{11}$$

In Eq. (11), min and max are the minimum and maximum values of respective K_P, K_I, λ, K_{P1}, K_D and μ parameters of FFOPI-FOPD controller. Imperialist competitive algorithm (ICA) is utilized for the purpose of optimization of the controller parameters. ICA is motivated by the imperialistic competition and socio-political progress of human beings. It provides high speed exploration and exploitation potential in searching the global optima. In this metaheuristic population-based evolutionary algorithm, each member is a country. ICA is written in .m file in MATLAB environment. For an N_{var} dimensional optimization problem, a country is a $1 \times N$ array given by

$$\text{country} = \left[P_1, P_2, \ldots, P_{N_{\text{var}}}\right] \tag{12}$$

$P_i s$ are the variables to be optimized. For FFOPI-FOPD AGC controller, the initial country is defined by

$$\text{country} = [K_P, K_I, \lambda, K_{P1}, K_D, \mu] \tag{13}$$

The studied single and multi-area power system models are implemented separately in Simulink of MATLAB. The cost function given by Eq. (9) is extracted as the output of Simulink models. To assess the cost functions, the inaugural matrixes of the countries are moved to the Simulink. Here, 500 countries are utilized to give 500 initial cost functions to be transferred later on to ICA .m file. Some of the powerful countries with lower cost function values are termed as imperialists. The rest countries constitute the colonies of imperialists. The imperialists with its colonies form some empires. The power of feeble empires will decline and stronger ones will swell successively as weaker empires lose their colonies to the stronger ones throughout the imperialistic competition. This competition is modelled by selecting the weakest colony of the weakest empire and attaching it to stronger empire, chosen on the basis of the competition among empires. The empires with more power have greater chances to acquire these colonies. During progress of the competition, weak empires will surrender their colonies and eventually, all the empires other than the most powerful one will crash. This one empire will administer all the colonies. Here, the power/cost and position of colonies and this unique empire are same. The first iteration will complete here. This operation ends up on attaining the stop condition of 100 iterations. The lowest cost function value among all the

Table 2
ICA optimized FPI/FFOPI/FFOPI-FOPD controller parameters.

Controller parameters	1-area non-reheat thermal system ($\Delta P_{D1}=0.01$ puMW)			2-area hydro-thermal system ($\Delta P_{D1}=0.01$ puMW)			3-area reheat thermal system ($\Delta P_{D1}=0.02$ puMW)		
	FPI	FFOPI	FFOPI-FOPD	FPI	FFOPI	FFOPI-FOPD	FPI	FFOPI	FFOPI-FOPD
K_P	0.8621	1.9058	1.8921	1.9395	2.4457	1.5124	0.9478	1.4412	1.5751
K_I	1.3969	1.2685	2.8142	1.8587	4.8554	4.9475	0.9145	1.5955	2.9047
λ	–	1.1358	1.1241	–	1.1057	0.9994	–	0.9879	0.9974
K_{P1}	–	–	1.8421	–	–	1.5738	–	–	1.7086
K_D	–	–	0.1114	–	–	0.4257	–	–	0.1215
μ	–	–	0.9874	–	–	0.7047	–	–	0.8084

An entry with "–" means not applicable.

cost functions will be chosen as the best one which results the optimum values of FFOPI-FOPD controller parameters. For more description, parameters, pseudo-codes and flow chart of ICA, the readers are advised to please refer authors' published works [22,29–31,36] and other prevalent works [13,14].

To compare the performance of different controllers presented therein, the values of IAE (7), ITAE (8) and ITSE (10) are also calculated. Following the same procedure stated above, to establish the supremacy of the proposed method, two more controllers i.e., fuzzy PI (FPI) and fuzzy FOPI (FFOPI) are also designed in the study. The parameters to be optimized in FPI and FFOPI are 2 (K_P and K_I) and 3 (K_P, K_I and λ), respectively. The optimized values of $K_P/K_I/\lambda/K_{P1}/K_D/\mu$ for 1-area non-reheat thermal, 2-area hydro-thermal and 3-area reheat thermal power system models for FPI/FFOPI/FFOPI-FOPD controller are given in Table 2.

5. Simulation results

5.1. 1-area non-reheat thermal power system

In this sub-section, the efficacy of the proposed controller is demonstrated for a 1-area non-reheat thermal power system model shown in Fig. 1 [4,8]. The data and nomenclature adopted are stated in Appendix. The simulation responses of change in frequency (ΔF) and change in power output (ΔP_g) of the system under a step load perturbation (SLP) of 20 MW i.e., 0.01 puMW applied at $t=0$ s are shown in Fig. 5(a) and (b). For comparison purpose, responses due to GA [4]/HS [8] tuned PID, GA tuned FOPID [4] and FPI/FFOPI controllers tuned in the study via ICA for the same system are also provided in Fig. 5(a) and (b). From the study of Fig. 5, it is prominent that FFOPI-FOPD controller outperforms optimized GA [4]/HS [8] PID, GA [4] FOPID and ICA FPI/FFOPI controllers with respect to less oscillations, peak undershoots (U_S), peak overshoots (O_S) and settling times (T_S). In defence of this finding, transient information about ΔF is displayed in Table 3. From the inspection of Table 3, it is inferred that the proposed FFOPI-FOPD controller is superior in terms of suppressing the deviation in frequency as the numerical values of U_S (–0.0036 Hz), O_S (0 Hz), T_S (1.25 s) and J_S like ISE ($3.14e^{-6}$), ITSE ($1.05e^{-6}$), IAE (0.0022) and ITAE (0.0039) are least with it compared to ICA FFOPI ($U_S=-0.0074$ Hz, $O_S=2.40e^{-4}$ Hz, $T_S=3.26$ s, ISE $=2.45e^{-5}$, ITSE $=1.93e^{-5}$, IAE $=0.0086$, ITAE $=0.0213$), ICA FPI ($U_S=-0.0107$ Hz, $O_S=2.65e^{-3}$ Hz, $T_S=4.25$ s, ISE $=5.39e^{-5}$, ITSE $=3.58e^{-5}$, IAE $=0.0109$, ITAE $=0.0214$),

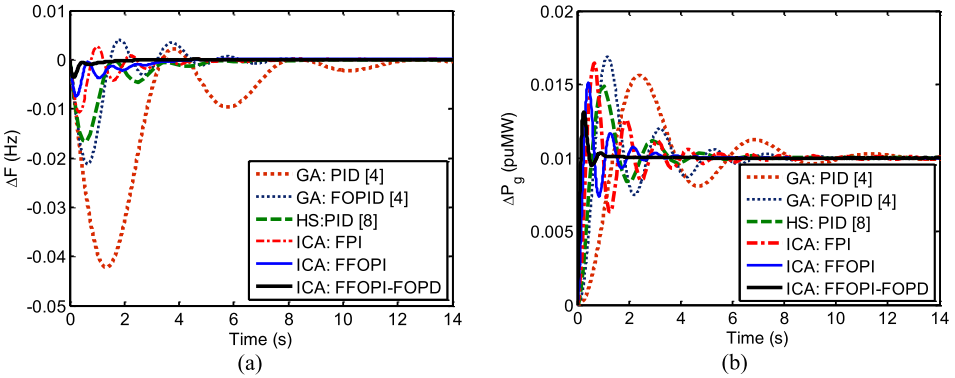


Fig. 5. Results of 1-area non-reheat thermal power system for SLP = 0.01 puMW. (a) Change in frequency response and (b) change in power generation response.

Table 3
Numerical values of T_S (tolerance band of ±0.0005), U_S/O_S and J_S (15 s) with 1-area non-reheat thermal power system at $\Delta P_D = 0.01$ puMW.

Controller type	T_S (s)	U_S (–ve) (Hz)	O_S (Hz)	J_S			
	ΔF	ΔF	ΔF	ISE	ITSE	IAE	ITAE
GA: PID [4]	14.88	0.0422	$2.22e^{-3}$	$2.80e^{-3}$	$4.72e^{-3}$	0.1087	0.3039
GA: FOPID [4]	7.16	0.0212	$4.05e^{-3}$	$3.27e^{-4}$	$2.67e^{-4}$	0.0276	0.0477
HS: PID [8]	5.03	0.0166	0	$1.97e^{-4}$	$1.57e^{-4}$	0.0213	0.0339
ICA: FPI	4.25	0.0107	$2.65e^{-3}$	$5.39e^{-5}$	$3.58e^{-5}$	0.0109	0.0214
ICA: FFOPI	3.26	0.0074	$2.40e^{-4}$	$2.45e^{-5}$	$1.93e^{-5}$	0.0086	0.0213
ICA: FFOPI-FOPD	**1.25**	**0.0036**	0	$\mathbf{3.14e^{-6}}$	$\mathbf{1.05e^{-6}}$	**0.0022**	**0.0039**

Bold designates best values.

HS PID [8] ($U_S = -0.0166$ Hz, $O_S = 0$ Hz, $T_S = 5.03$ s, ISE = $1.97e^{-4}$, ITSE = $1.57e^{-4}$, IAE = 0.0213, ITAE = 0.0339), GA FOPID [4] ($U_S = -0.0212$ Hz, $O_S = 4.05e^{-3}$ Hz, $T_S = 7.16$ s, ISE = $3.27e^{-4}$, ITSE = $2.67e^{-4}$, IAE = 0.0276, ITAE = 0.0477) and GA PID [4] ($U_S = -0.0422$ Hz, $O_S = 2.22e^{-3}$ Hz, $T_S = 14.88$ s, ISE = $2.80e^{-3}$, ITSE = $4.72e^{-3}$, IAE = 0.1087, ITAE = 0.3039). In the last row of the relevant tables, the results of attention are **bold** faced.

5.2. 2-area hydro-thermal system

To signify the talent of ICA optimized FFOPI-FOPD controller to cope with multi-area systems, the approach is extended to a hydro-thermal 2-area 2-unit power system model shown in Fig. 2 [3]. Every control area has 1 non-reheat thermal and 1 mechanical governor based hydro units. The system results with the suggested controller, ICA tuned FPI/FFOPI controller and some other recently published best asserted controllers like GA [3]/QODSA [9] tuned PI and BBBO [5]/hFA-PS [10]/hGGSA-PS [11] tuned PID under 1% SLP at $t = 0$ s in area-1 for deviation in frequency of area-1 (ΔF_1), deviation in frequency of area-2 (ΔF_2), deviation in tie-line power (ΔPtie_{12}) and ACE of area-1 (ACE_1) are given in Fig. 6(a)–(d). By inspection of these figures, it is observed that the system results due to FFOPI-FOPD controller are amazingly better than the other controllers optimized for the same system in terms of

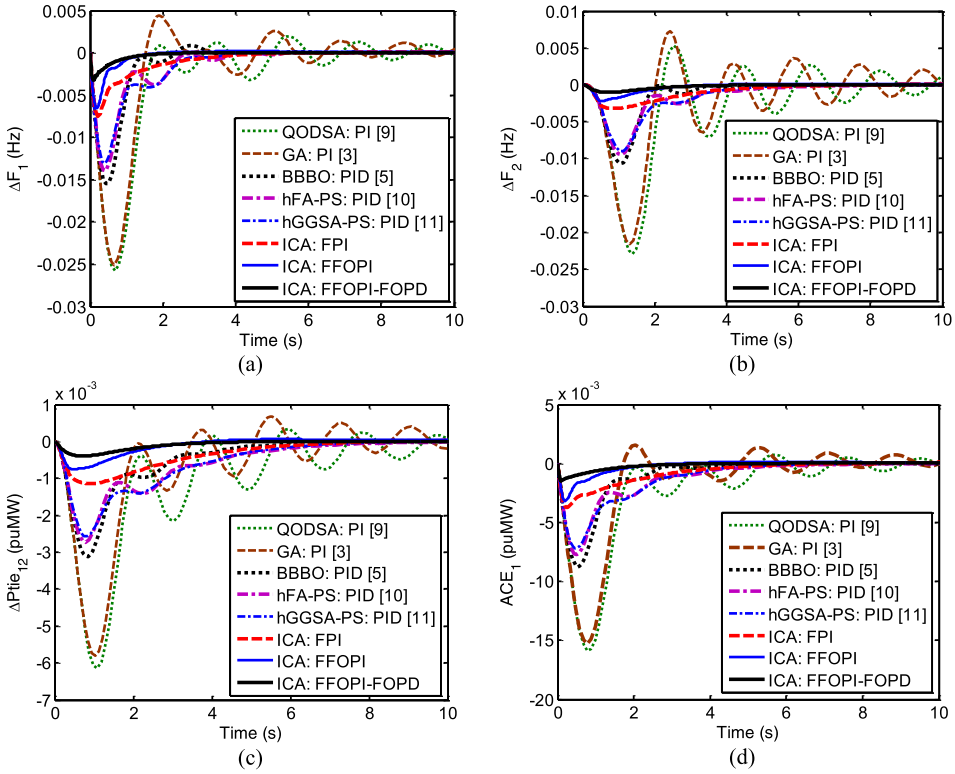


Fig. 6. Results of 2-area hydro-thermal power system for SLP = 0.01 puMW in area-1. (a) Change in area-1 frequency, (b) change in area-2 frequency, (c) change in tie-line power and (d) change in area-1 area control error.

fast and oscillations free responses showing least deviations from the desired 0 steady state value with minimal settling time. The numerical values of $T_S/U_S/O_S$ of $\Delta F_1/\Delta F_2/\Delta \text{Ptie}_{12}$ response and ISE/ITSE/IAE/ITAE with the proposed ICA optimized FFOPI-FOPD controller and other controllers are given in Table 4. It is evident from Table 4 that minimum J_S (ISE = $4.68e^{-6}$, ITSE = $2.74e^{-6}$, IAE = 0.0051, ITAE = 0.0098), T_S ($\Delta F_1 = 1.12$ s, $\Delta F_2 = 1.89$ s, $\Delta \text{Ptie}_{12} = 0.72$ s) and U_S ($\Delta F_1 = -0.0031$ Hz, $\Delta F_2 = -0.0010$ Hz, $\Delta \text{Ptie}_{12} = -0.00039$ puMW) are obtained with the proposed controller compared to GA [3]/QODSA [9]: PI, BBBO [5]/hFA-PS [10]/hGGSA-PS [11]: PID and ICA: FPI/FFOPI controllers. Hence, ICA FFOPI-FOPD controller show superior performance in AGC.

5.3. 3-area reheat thermal system

Additionally, competence of the proposed approach is validated on a 3-area unequal reheat thermal power system shown in Fig. 3 [2,7]. The control area power rating of 1/2/3 area is 2000 MW/5000 MW/8000 MW, respectively. The system results with the suggested controller, ICA: FPI/FFOPI controller and some other recently published controllers like KHA: PI/PID [7], optimal: PI [2] and GWO: PID [6] under 2% SLP at $t = 0$ s in area-1 for $\Delta F_1/\Delta F_2/\Delta F_3/\Delta \text{Ptie}_1/\Delta \text{Ptie}_2/\Delta \text{Ptie}_3$ are shown in order in Fig. 7(a)–(f). The system performance of $\Delta F_1/\ \Delta F_2/\Delta F_3$ response in terms of numerical values of $O_S/U_S/T_S/J_S$ is illustrated

Table 4
Numerical values of T_S (tolerance band of ±0.0005), U_S/O_S and J_S (15 s) with 2-area hydro-thermal power system at $\Delta P_{D1} = 0.01$ puMW.

Controller type	T_S (s)			U_S (–ve) (Hz)		U_S (–ve) (puMW)	O_S (Hz)		O_S (puMW)	J_S			
	ΔF_1	ΔF_2	ΔPtie_{12}	ΔF_1	ΔF_2	ΔPtie_{12}	ΔF_1	ΔF_2	ΔPtie_{12}	ISE	ITSE	IAE	ITAE
QODSA: PI [9]	11.46	14.22	7.17	0.0258	0.0229	0.00615	$1.99e^{-3}$	$5.32e^{-3}$	$3.44e^{-4}$	$1.02e^{-3}$	$1.31e^{-3}$	0.0910	0.2644
GA: PI [3]	15.84	18.55	6.49	0.0251	0.0215	0.00581	$4.39e^{-3}$	$7.32e^{-3}$	$6.73e^{-4}$	$9.03e^{-4}$	$1.18e^{-3}$	0.0875	0.2772
BBBO: PID [5]	3.65	4.45	3.16	0.0155	0.0106	0.00311	$8.63e^{-4}$	0	0	$2.23e^{-4}$	$1.67e^{-4}$	0.0302	0.0533
hFA-PS: PID [10]	3.74	4.80	4.21	0.0139	0.0094	0.00272	0	0	0	$1.91e^{-4}$	$1.69e^{-4}$	0.0341	0.0570
hGGSA-PS: PID [11]	3.39	4.53	4.12	0.0131	0.0089	0.00257	0	0	0	$1.88e^{-4}$	$1.77e^{-4}$	0.0356	0.0645
ICA: FPI	3.36	4.44	3.15	0.0075	0.0031	0.00114	0	0	0	$4.88e^{-5}$	$4.99e^{-5}$	0.0214	0.0532
ICA: FFOPI	1.26	2.19	1.41	0.0065	0.0022	0.00076	0	0	0	$1.77e^{-5}$	$1.04e^{-5}$	0.0104	0.0288
ICA: FFOPI-FOPD	**1.12**	**1.89**	**0.72**	**0.0031**	**0.0010**	**0.00039**	0	0	0	$\mathbf{4.68e^{-6}}$	$\mathbf{2.74e^{-6}}$	**0.0051**	**0.0098**

Bold designates best values.

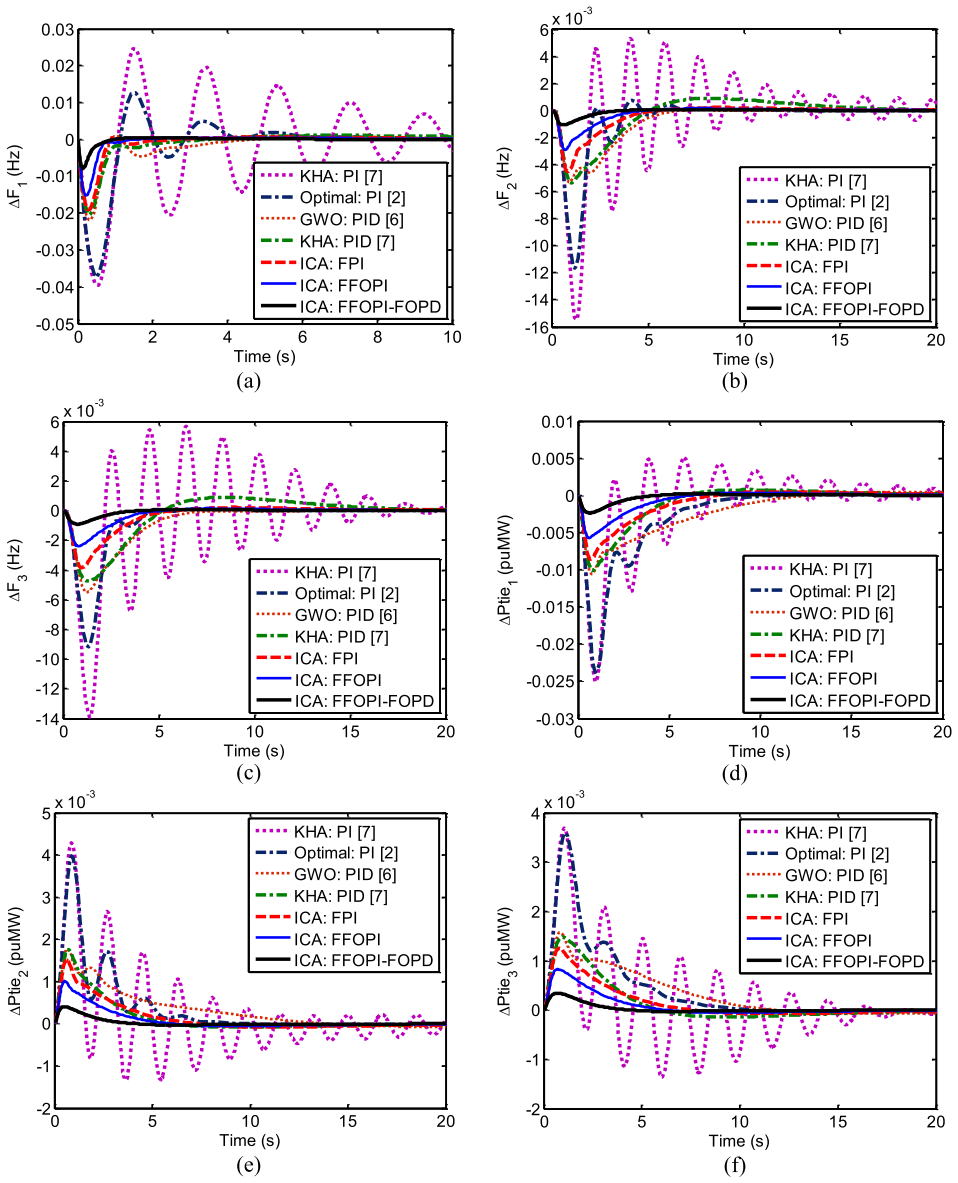


Fig. 7. Results of 3-area reheat thermal power system for SLP = 0.02 puMW in area-1. (a) Change in area-1 frequency, (b) change in area-2 frequency, (c) change in area-3 frequency, (d) change in tie-line-1 power, (e) change in tie-line-2 power and (f) change in tie-line-3 power.

in Table 5. On critical study of Fig. 7(a)–(f) and Table 5, it is recognized that the system dynamic performance offered by the proposed approach is better in every respect of $O_S/U_S/T_S/J_S$ than prevalent KHA: PI/PID [7]/optimal: PI [2]/GWO: PID [6] and designed ICA: FPI/FFOPI control methods for AGC of 3-area reheat thermal system.

Table 5
Numerical values of T_S (tolerance band of ±0.0005), U_S/O_S and J_S (15 s) with 3-area reheat thermal power system at $\Delta P_{D1} = 0.02$ puMW.

Controller type	T_S (s)			U_S (–ve) (Hz)		U_S (–ve) (puMW)	O_S (Hz)		O_S (puMW)	J_S			
	ΔF_1	ΔF_2	ΔF_3	ΔF_1	ΔF_2	ΔF_3	ΔF_1	ΔF_2	ΔF_3	ISE	ITSE	IAE	ITAE
KHA: PI [7]	24.83	22.61	16.27	0.0397	0.0179	0.0169	0.02460	0.0053	0.0056	0.00274	0.00672	0.2695	1.2100
Optimal: PI [2]	7.68	4.39	4.09	0.0374	0.0170	0.0091	0.01270	0.0007	0	0.00136	0.00133	0.1273	0.2596
GWO: PID [6]	8.44	5.11	5.37	0.0219	0.0052	0.0055	0.00090	0	0	0.00043	0.00062	0.0965	0.2939
KHA: PID [7]	11.90	12.49	12.61	0.0204	0.0053	0.0047	0.00106	0.0009	0.0008	0.00036	0.00049	0.0901	0.3504
ICA: FPI	6.35	3.87	4.01	0.0191	0.0045	0.0038	0.00059	0	0	0.00023	0.00020	0.0583	0.1718
ICA: FFOPI	4.01	3.10	3.26	0.0153	0.0029	0.0023	0.00051	0	0	0.00011	0.00008	0.0369	0.1017
ICA: FFOPI-FOPD	**0.76**	**1.65**	**1.79**	**0.0078**	**0.0010**	**0.0009**	**0.00045**	0	0	**0.00002**	**0.00001**	**0.0139**	**0.0332**

Bold designates best values.

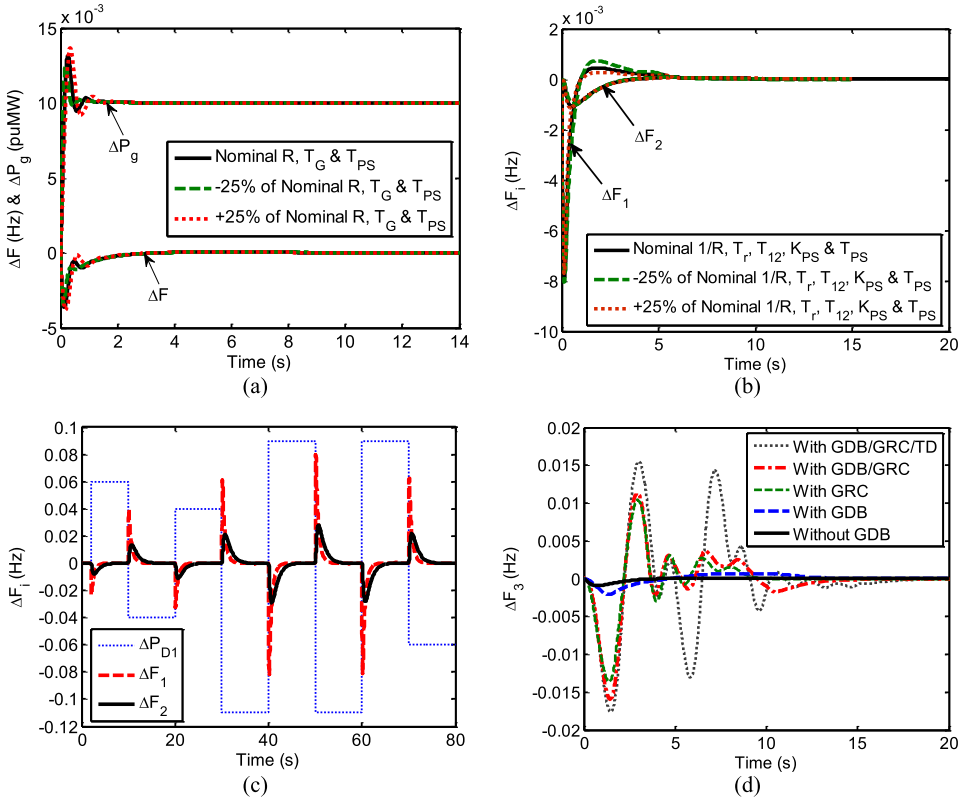


Fig. 8. Results under sensitivity analysis. (a) Change in area frequency and generation for SLP = 0.01 puMW of 1-area non-reheat thermal system, (b) change in area frequencies at SLP = 0.02 puMW in area-1of 3-area reheat thermal system, (c) change in area frequencies for random SLP in area-1 of 2-area hydro-thermal power system and (d) change in area-3 frequency at SLP = 0.02 puMW in area-1 of 3-area reheat thermal system.

6. Sensitivity analysis

Here, a sensitivity analysis [1,2,9,10,21–26,28–31,33–41] is carried out to check the robustness of proposed FFOPI-FOPD controller and ICA. In this work, 3 types of robustness studies are performed. In 1st type, robustness of FFOPI-FOPD controller is investigated under wide changes in the system parameters, in 2nd type, robustness is confirmed at variable load demands and in 3rd type, robustness is analysed in the presence of physical constraints. The designed controller should work robustly devoid of substantial decline in system performance under these conditions.

Fig. 8(a) shows $\Delta F/\Delta P_g$ response at nominal/±25% change in nominal $R/T_G/T_{PS}$ parameter for 1-area non-reheat thermal power system. Fig. 8(b) shows $\Delta F_1/F_2$ response at nominal/±25% variation in nominal $(1/R)/T_r/T_{12}/K_{PS}/T_{PS}$ parameter for 3-area reheat thermal power system. The review of Fig. 8(a) and (b) indicates that the responses for both the systems due to FFOPI-FOPD controller are more or less same without significant decline under ±25% alteration in the system parameters. Thus, the proposed FFOPI controller displays robust behaviour under wide changes in parameters of the systems.

Next, the robustness of FFOPI-FOPD controller is tested under random load demands and physical constraint like generation rate constraint (GRC), governor deadband (GDB) and time delay (TD) in view of real time practical conditions and power system nature. If controller is not robust, the system will become unstable under random load demands and GRC/GDB/TD. Fig. 8(c) shows $\Delta F_1/F_2$ response of 2-area hydro-thermal system with random SLP in area-1. Fig. 8(d) shows ΔF_3 response of 3-area reheat thermal system without acknowledging GRC, with GDB, with GRC, with GRC/GDB and with combined GRC/GDB/TD. The value of closed loop type GRC used in turbine is ± 0.0017 puMW/s, GDB value selected is ± 0.006 at the output signal of speed governor and 50 ms transport type TD is used in the ACE signal and after $(1/R)$ block in each area of 3-area system. Fig. 8(c) and (d) clearly reveals that the FFOPI-FOPD controller works healthily as the system responses are stable under random SLPs and GRC/GDB/TD nonlinearity. However, system performance in the presence of GRC/GDB/TD degrades significantly. The deterioration of the system performance takes place in the order of without GDB → with GDB → with GRC → with GRC/GDB → with GRC/GDB/TD. Thus, it is said that ICA optimized parameters of FFOPI-FOPD controller obtained at nominal conditions are robust and need not be re-optimized under alterations in the system parameters, random SLPs and physical constraints.

7. Conclusion and future scope

A new FFOPI-FOPD controller is proposed for the effective solution of AGC problem in single-area and interconnected multi-area electric power systems. ICA is employed to optimize the parameters of FFOPI-FOPD controller using ISE error criterion. The superiority of the controller is demonstrated by contrasting its responses with various optimal controllers which appeared recently in the literature and ICA tuned FPI/FFOPI controller designed in the study for 1-area thermal, 2-area hydro-thermal and 3-area thermal power systems. Investigations reveal that the proposed controller furnishes superior dynamic performance compared to other methods in terms of minimum values of various error criteria and undershoots/overshoots/settling times of change in ACE/frequency/generation/tie-line power response following a sudden load demand in a control area. The sensitivity analysis authenticates robustness of ICA optimized FFOPI-FOPD controller gains to ample changes in the system parameters, random load demands and system nonlinearities. Hence, the proposed approach is well suitable for AGC problem solution of power systems. Over the existing methods, it has various advantages such as simplicity of design, easy to implement, fast/smooth responses, least oscillatory outcome, economic, enhanced equipment life and robust behaviour.

As scope of future work, to reduce conservatism, there is still room in the present study for further improvement in AGC of power system. To enhance the performance, in future, the input SFs of the proposed controller may be optimized in addition to output SFs. The advocated controller may be optimized via some other latest optimization techniques. The proposed approach may be implemented on some other real-world traditional and deregulated power system AGC problems. Some other control structure like fuzzy FO cascade controller, fuzzy tilt controller, fuzzy tilt cascade controller etc. may be designed for the studied systems. To appraise the impact of renewable energy sources (RES) and energy storage systems (ESS), RES/ESS may be integrated with the studied models in future course of work. By adopting the mentioned approaches, the conservatism of the study may be minimized to a great extent.

Appendix: Nomenclature and system data [2–4,7]

i	= subscript referring to area-i ($i=1, 2, 3$)	T_{RHi}	= 48.7 s; hydro turbine speed governor transient droop time constant
α_{12}	= –1 ($\alpha_{12}=-2/5$, $\alpha_{13}=-2/8$, $\alpha_{23}=-5/8$, for 3-area system); area size ratio	T_{Ri}	= 5 s; hydro turbine speed governor reset time
F0	60 Hz; nominal system frequency	T_{GHi}	= 0.513 s; hydro turbine speed governor main servo time
R_i	= 2.4 Hz/puMW ($R_1=2$ for 2-area hydro-thermal system, $R=3$ for 1-area thermal system); speed governor regulation parameter	T_{12}	= 0.0707 puMW/rad for 2-area hydro-thermal system and $T_{12}=0.0866$ puMW/rad for 3-area system); tie-line synchronizing coefficient
β_i	= 0.425; frequency response characteristic	T_{Wi}	= 1 s; water starting time constant
K_{PSi}	= 120 ($K_{PS}=100$ for 1-area system); power system gain	α_{Ti}/α_{Hi}	0.5; participation factor in 2-area hydro-thermal power system
T_{PSi}	= 20 s; power system time constant	ACE_i	area control error
T_{Gi}	= 0.08 s ($T_G=0.4$ for 1-area system); speed governor time constant	ΔP_{Di}	incremental change in area load power demand
T_{Ti}	= 0.3 s ($T_T=0.5$ for 1-area system); thermal turbine time constant	ΔP_{gi}	incremental change in area power generation
K_{ri}	= 0.5; reheater gain	ΔF_i	incremental change in the area frequency
T_{ri}	= 10 s; reheater time constant	ΔP_{Ci}	incremental change in the speed changer position

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